

**PART A**  
(25 marks)

1. Evaluate each of the following limits

(a)  $\lim_{x \rightarrow 2} \frac{\sqrt{x+7} - 3}{x - 2}$  (3 marks)

(b)  $\lim_{x \rightarrow 3^-} \frac{9 - x^2}{|x - 3|}$  (3 marks)

(c)  $\lim_{x \rightarrow -\infty} \frac{\sqrt{x^2 + 25}}{x - 2}$  (4 marks)

2. (a) Given  $f(x) = \frac{1}{2x+5}$

By using first principle of derivative, find  $f'(x)$  (5 marks)

(b) Find the derivative of  $f(x) = \frac{1 + \sqrt{x}}{1 - \sqrt{x}}$ . (5 marks)

3. Given the curve  $f(x) = x^3 + 3x^2 - 24x + 20$ . Use the 2<sup>nd</sup> derivative test to determine the maximum and minimum points of  $f(x)$ .

(5 marks)

## PART B

(75 marks)

1. It is given that  $w = a + bi$ ,  $|w| = 4$  and  $\arg w = \frac{5\pi}{6}$ .
- (a) Find the values of  $a$  and  $b$ . (4 marks)
- (b) Find  $|zw|$ , if  $z = a - \frac{b}{2}i$ . Hence, find the  $z$  in the polar form. (5 marks)
2. Find the set of value which satisfy  $\left|\frac{1}{x}\right| \geq \frac{3}{2x-1}$ . (11 marks)
3. A function is defined by  $g(x) = \sqrt{2 - \frac{x}{2}}$ .
- (a) Show that  $g(x)$  is one-to-one function algebraically. (2 marks)
- (b) Find  $f(x)$ , if  $f(x) = g^{-1}\left(\frac{x}{6}\right)$ . (5 marks)
- (c) State the domains and ranges of both function  $g(x)$  and  $g^{-1}(x)$ . (2 marks)
- (d) Sketch the graphs of  $g(x)$  and  $g^{-1}(x)$  on the same coordinates axes. (3 marks)
4. Given the function
- $$r(x) = 2e^{5x} - 1 \text{ for all real values of } x \text{ and}$$
- $$s(x) = \frac{1}{5} \ln\left(\frac{x+1}{2}\right) \text{ for } x > -1.$$
- Find  $(r \circ s)(x)$  and  $(s \circ r)(x)$ . Hence, state the conclusion about the results. (6 marks)
5. (a) The function  $m$  is defined as  $m(x) = \frac{3x}{\sqrt{4x^2 - 25}}$ .
- (i) State the domain of  $m$ . (2 marks)
- (ii) Find the vertical asymptotes of  $m$ . (3 marks)
- (iii) State the horizontal asymptotes of  $m$ , if exist. (3 marks)

- (b) The function  $k$  is defined as

$$k(x) = \begin{cases} Mx^2 - 6 & ; \quad x \leq -\frac{5}{4} \\ Nx + M & ; \quad -\frac{5}{4} < x < 7 \\ 25 & ; \quad x \geq 7. \end{cases}$$

Determine the values of  $M$  and  $N$  if  $k$  is continuous at  $x = -\frac{5}{4}$  and  $x = 7$ .

(4 marks)

6.

- (a) Find  $\frac{dy}{dx}$  of  $y = \ln\left(\frac{\ln x}{\sqrt{x}}\right)$  (4 marks)

- (b) Parametric equations of a curve are given by  $x = \frac{2t^4 - 1}{t^3}$  and  $y = 2t + \frac{1}{t^3}$ , where  $t$  is a non-zero parameter.

- (i) Find  $x - y$  and  $(x + y)^3$  in terms of  $t$ . Hence, show that  $(x - y)(x + y)^3 = -128$ . (6 marks)

- (ii) Calculate the value of  $\frac{dy}{dx}$  when  $t = 1$  (6 marks)

7.

A cuboid has a square base of sides  $x$  cm. the height of the cuboid is twice the length of the side of its base. Given that the total surface area is  $250\text{cm}^2$  and  $x$  increase at the rate of  $0.05\text{cms}^{-1}$ . Find the rate of change of the volume of the cuboid.

(9 marks)

END OF QUESTIONS PAPER

Suggestion answers:

## Part A

1. (a)  $\frac{1}{6}$  (b) 6 (c) -1

2. (a)  $\frac{-2}{(2x+5)^2}$  (b)  $\frac{1}{\sqrt{x}(1-\sqrt{x})^2}$

3. (2,-8) min point / (-4,100) max point

## Part B

1. (a)  $a = -2\sqrt{3}$  ;  $b = 2$

(b)  $|zw| = 4\sqrt{13}$  ;

$z = \sqrt{13}(\cos(-2.8606 \text{ rad.}) + i \sin(-2.8606 \text{ rad.}))$

2.  $\left\{x : x < 0 \text{ or } 0 < x < \frac{1}{2}\right\}$

3. (a) Shown

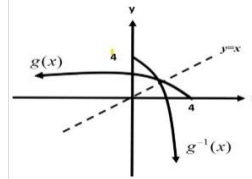
(b)  $f(x) = 4 - \frac{x^2}{18}$

(c)

$D_g = (-\infty, 4] = R_{g^{-1}}$

$R_g = [0, \infty) = D_{g^{-1}}$

(d)



4.  $(r \circ s)(x) = x$  ,  $(s \circ r)(x) = x$  ,  $r(x)$  and  $s(x)$  are inverse of each others

5. (a) (i)  $D_m = \left(-\infty, -\frac{5}{2}\right) \cup \left(\frac{5}{2}, \infty\right)$

(ii)  $x = -\frac{5}{2}$  and  $x = \frac{5}{2}$

(iii)  $y = -\frac{3}{2}$  and  $y = \frac{3}{2}$

(b)  $M = 4$  ,  $N = 3$

6. (a)  $\frac{1}{x \ln x} - \frac{1}{2x}$

(b) (i) Shown

(ii)  $-\frac{1}{5}$

7. 7.5